BBT-059, a PEGylated Interleukin-11 Analog, Confers Radioprotection Against Low- and High-LET Radiation

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Purpose: Nuclear detonations pose a unique threat due to high-LET mixed-field (neutron/gamma) radiation, which causes more complex damage than standard gamma rays. This study evaluates BBT-059, a long-acting PEGylated interleukin-11 (IL-11) analog, as pre-exposure radioprotectant against hematopoietic sub-syndrome of acute radiation syndrome (H-ARS).

Methods: Male and female C57BL/6 mice (11 to14 weeks old) were administered a single subcutaneous dose of BBT-059 (0.3 mg/kg) or vehicle (formulation buffer) 12 hours prior to irradiation. Mice were exposed to 0.6 Gy/min at the AFRRI High-level Cobalt-60 facility (low-LET gamma) or the TRIGA reactor facility (high-LET mixed-field) to simulate fallout and detonation scenarios, respectively.
Thirty-day survival was analyzed via Kaplan-Meier plots. Hematopoietic recovery was assessed through peripheral blood counts, serum biomarker assays, and flow cytometry of bone marrow populations. Histopathological analysis (hematoxylin and eosin staining) and RT-PCR profiler arrays for pathway analysis were performed on collected tissues. Statistical significance was determined using Mantel-Cox and Fisher's exact tests for survival, and 2-way ANOVA for other parameters.

Results: Pre-treatment with BBT-059 significantly improved 30-day survival in mice across both radiation qualities. Treated mice exhibited accelerated reconstitution of bone marrow progenitor cells and peripheral blood counts. Mechanistic analysis suggests BBT-059 mitigates H-ARS by stimulating early-stage hematopoietic stem cell recovery.

Conclusions: A single pre-exposure dose of BBT-059 provides potent, broad-spectrum protection against diverse radiation types, offering a critical capability for enhancing personnel survival and operational readiness.

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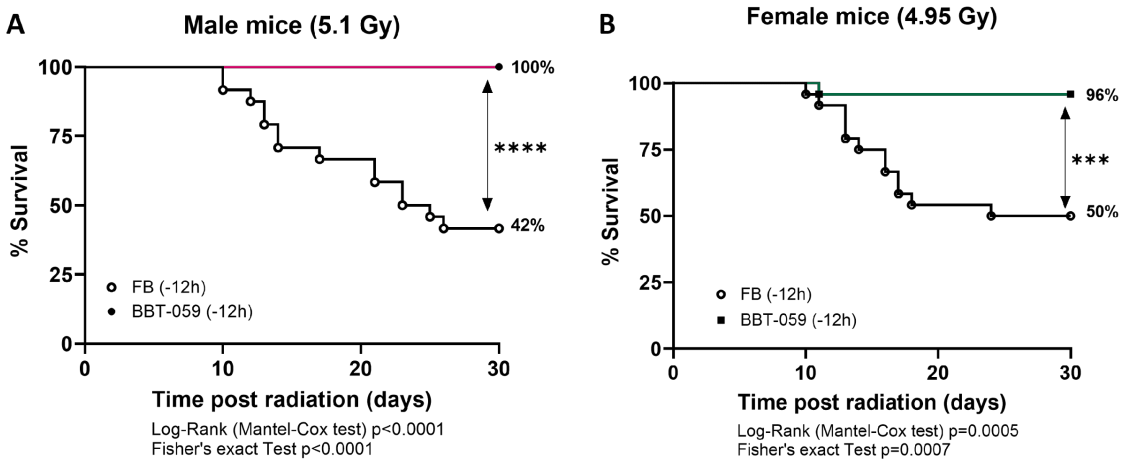

Figure 1. Radioprotective efficacy of BBT-059 against high-LET radiation. C57BL/6 mice (A: male, B: female) were treated with BBT-059 or formulation buffer (FB) and exposed to mixed-field

(neutron/gamma) radiation. Pre-treatment with BBT-059 significantly increased post-irradiation survival compared to the formulation buffer (FB) group. *$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$, ****$p \leq 0.0001$.